

String Matching

Time limit: 1.00 s

Memory limit: 512 MB

Given a string and a pattern, your task is to count the number of positions where the pattern occurs in the string.

Input

The first input line has a string of length n , and the second input line has a pattern of length m . Both of them consist of characters a–z.

Output

Print one integer: the number of occurrences.

Constraints

- $1 \leq n, m \leq 10^6$

Example

Input	Output
saippuakauppias pp	2